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MODERN DELAY-TOLERANT EMAIL

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Abstract

Abstract content

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Contents

Abstract	i
1 Introduction	1
1.1 Introduction	1
1.2 Delay/Disruption-Tolerant Networking	1
1.2.1 Concept and Motivation	1
1.2.2 DTN Architecture and Bundle Layer	2
1.2.3 DTN Nodes, Regions, and Security	2
1.2.4 Routing, Scheduling, and Buffer Management	3
1.3 Internet Email Architecture	3
1.3.1 Service Model and Roles	3
1.3.2 Protocol Suite: SMTP, POP, IMAP, MIME	4
1.3.3 Security and Anti-Abuse Mechanisms	4
1.4 Comparison of DTN and Email Architectures	5
1.5 Bundle Protocol in Interplanetary DTN	5
1.5.1 Interplanetary Architecture and Gateways	6
1.5.2 Naming, DNS, and BP Node Addressing	6
1.5.3 Interplanetary Email over the Bundle Protocol	6
1.6 Interplanetary Overlay Network (ION)	7
1.6.1 System Architecture	7
1.6.2 Applications and Deployment	8
1.7 Postfix	8

1.8	Dovecot	8
1.9	Thunderbird	8
1.10	Virtual Machines (VMs)	9
1.11	ION DTN	9
2	Literature Review	10
3	System Design	11
4	Implementation	12
5	Results and Discussion	13
6	Conclusions and Future Work	14

List of Figures

List of Tables

1 | Introduction

1.1 Introduction

Delay/Disruption-Tolerant Networking (DTN) and Internet email are both store-and-forward communication systems that enable asynchronous message delivery across heterogeneous networks (1, 2). DTN generalizes the store-and-forward concept as a network-layer or overlay architecture for challenged environments, whereas email is a specific application-layer service built on top of relatively well-connected IP networks.(1, 2)

1.2 Delay/Disruption-Tolerant Networking

1.2.1 Concept and Motivation

DTN targets “challenged” networking environments characterized by intermittent connectivity, long or variable delays, high error rates, and constrained resources such as power and storage.(1, 3) In such environments, classical TCP/IP assumptions about continuous end-to-end paths and low round-trip time do not hold, which requires a different architectural approach.(1)

Typical DTN scenarios include deep-space communication, rural and remote connectivity, ad hoc vehicular and mobile networks, and disaster-response deployments where infrastructure is damaged or overloaded.(1, 3) The primary objective is to provide eventual data delivery despite disruptions, using persistent

storage and opportunistic forwarding across time-varying connectivity graphs.(1, 4)

1.2.2 DTN Architecture and Bundle Layer

The canonical DTN architecture introduces a new overlay protocol layer, commonly called the bundle layer, which resides above region-specific transports (e.g., TCP, UDP, Licklider Transmission Protocol) and below applications.(1, 3) Applications generate protocol data units called bundles, which encapsulate payload data together with endpoint identifiers, lifetime, priority, and optional security blocks.(1, 5)

Each DTN node maintains persistent storage for bundles and performs a store-carry-forward operation: bundles are stored locally, carried while the node is mobile or idle, and forwarded when a suitable contact becomes available.(1, 6)

Convergence-layer adapters map bundle operations onto specific underlying transports (for example, the DTN TCP Convergence-Layer Protocol) so that the bundle layer is insulated from link heterogeneity.(5)

1.2.3 DTN Nodes, Regions, and Security

The architecture partitions the overall system into regions, such as terrestrial IP networks, satellite segments, and deep-space links, which may employ different lower-layer protocols and routing schemes.(3) The bundle layer provides interoperability across these regions using endpoint identifiers and late binding, allowing applications to communicate without being aware of underlying heterogeneity.(3, 6)

Security is addressed using dedicated bundle security mechanisms that provide authentication, integrity protection, and confidentiality over multi-hop, disruption-prone paths.(3) Because DTN nodes may store data for long periods and may replicate messages to improve delivery probability, the security design also considers access control, resistance to resource exhaustion, and key management

across regions.(4, 6)

1.2.4 Routing, Scheduling, and Buffer Management

Routing in DTNs differs fundamentally from traditional IP routing because an end-to-end path between source and destination may not exist at any given time.(4, 6) Instead, routing algorithms operate over time-varying or probabilistic contact graphs, taking into account predicted or observed node mobility and contact opportunities.(6)

Existing DTN routing schemes can be broadly classified by the degree of replication and the information they exploit: epidemic and multi-copy protocols maximize delivery probability at the cost of bandwidth and buffer usage, while single-copy or quota-based schemes trade delivery ratio for reduced overhead.(4, 6) Since contacts are intermittent and buffers are limited, joint scheduling and buffer management policies are required to decide which bundles to transmit or drop according to priorities, deadlines, and resource constraints.(6)

1.3 Internet Email Architecture

1.3.1 Service Model and Roles

Internet email is an asynchronous application-layer service that delivers messages between users identified by email addresses of the form `local-part@domain`.(2, 7) The system is decomposed into functional roles including Message User Agents (MUAs), Message Submission Agents (MSAs), Message Transfer Agents (MTAs), Message Delivery Agents (MDAs), and Mail Access Agents.(2)

MUAs (such as desktop clients or webmail front-ends) are responsible for message composition, display, and user interaction, but they offload submission, relay, and storage to server-side agents.(2) MSAs accept messages from MUAs, apply initial checks, and hand them to MTAs, which relay messages between domains until they

reach an MDA that deposits them into per-user mailboxes.(2, 7)

1.3.2 Protocol Suite: SMTP, POP, IMAP, MIME

The Simple Mail Transfer Protocol (SMTP) defines the submission and relay of messages between MSAs and MTAs, including envelope commands and response codes for reliable transfer (7). Originally designed for clear text operation and implicit trust between servers, SMTP has been extended with authentication and encryption mechanisms such as SMTP AUTH and STARTTLS.(7)

For mailbox access, the Post Office Protocol version 3 (POP3) supports simple download-and-delete retrieval, while the Internet Message Access Protocol (IMAP) offers folder hierarchies, server-side search, and synchronization across multiple clients and devices.(8) At the message format level, the Internet Message Format standard and Multipurpose Internet Mail Extensions (MIME) specify headers, structure, multiple body parts, and attachments, enabling rich content and internationalization.(9)

1.3.3 Security and Anti-Abuse Mechanisms

Traditional email architecture did not include end-to-end confidentiality, strong sender authentication, or robust abuse controls, making it vulnerable to spoofing, spam, and phishing (2). In practice, deployments mitigate these weaknesses by combining transport-layer security (TLS), server and user authentication, and extensive content-based filtering, often using machine learning to classify spam and malicious messages.(2, 10)

Domain-level authentication and policy mechanisms such as Sender Policy Framework (SPF), DomainKeys Identified Mail (DKIM), and DMARC enable domains to assert which hosts may send mail on their behalf and to specify handling rules for authentication failures.(2) Additional secure application architectures have been proposed that overlay existing email infrastructure with stronger identity,

integrity, and legal guarantees, while preserving compatibility with standard protocols.(10)

1.4 Comparison of DTN and Email Architectures

Both DTN and Internet email rely on asynchronous, store-and-forward operation, with intermediate nodes temporarily buffering data before forwarding it to the destination.(1, 2) However, DTN is conceived as a general-purpose network-layer architecture for environments with extreme delay and disruption, while email is a specific application protocol suite designed for relatively stable IP networks (1, 7).

DTN introduces a unified bundle layer to integrate heterogeneous underlying networks and explicitly models time-varying connectivity and contact opportunities (1, 3). On the contrary, the email architecture builds on existing IP transport and focuses on application-level roles, standardized message formats, and security mechanisms for addressing and content-level

1.5 Bundle Protocol in Interplanetary DTN

The Bundle Protocol (BP) is the core network protocol of Delay/Disruption-Tolerant Networking (DTN), designed to support long-delay and intermittently connected paths using store-and-forward transfer of application data units called bundles (11). In interplanetary scenarios, BP is typically deployed as an overlay over one or more convergence layers (such as TCP, LTP, or space link protocols), providing persistent storage and late binding of next hops so that disruptions that would break conventional end-to-end transports do not prevent eventual delivery (11). This overlay role allows BP to act as a unifying backbone across heterogeneous regional networks, particularly where deep-space links have radically different characteristics from local terrestrial segments (11).

1.5.1 Interplanetary Architecture and Gateways

Proposed interplanetary network architectures generally assume that local networks on Earth, the Moon, and Mars are IP-based, while deep-space and interplanetary trunks use BP to interconnect these IP “islands” (12). IP/BP gateways terminate local IP sessions, encapsulate application-layer traffic (for example HTTP or SMTP) into bundle payloads, and forward those bundles over scheduled BP contacts so that remote gateways can reconstruct the original IP traffic within their own local environments (12). This architecture enables reuse of existing Internet applications without requiring native end-to-end IP connectivity across long-delay space links (12).

1.5.2 Naming, DNS, and BP Node Addressing

To integrate BP-based interplanetary connectivity with name resolution, an interplanetary DNS model has been proposed in which each planetary body operates its own DNS root system and associated top-level domains (13). In this model, DNS records for those TLDs resolve primarily to local IP/BP gateways rather than directly reachable off-world IP addresses, and can include resource records that map domain names to BP node identifiers as well as MX records pointing to interplanetary mail gateways (13). DNSSEC can still be applied with distinct trust anchors per world, while most DNS traffic remains local and only application data is carried over BP between planets (13).

1.5.3 Interplanetary Email over the Bundle Protocol

Email is a prominent application driving the deployment of DTN and BP in interplanetary contexts, where entire SMTP exchanges or packeted email messages are encapsulated as bundle payloads and transported between IP networks on different planetary bodies (12, 14). One proposal describes several deployment models where Mail Transfer Agents (MTAs) cooperate with Bundle Protocol Agents

(BPAs) to move email between Earth and remote BP nodes, including cases in which the remote node has only BP connectivity and relies on a local shim to provide SMTP semantics to users or applications (12). A complementary specification defines a concise encapsulation based on Batch SMTP media types and a dedicated BP service number, allowing conventional MTAs at each end to operate unchanged while the interplanetary hop is handled solely via BP (14).

1.6 Interplanetary Overlay Network (ION)

The Interplanetary Overlay Network (ION) is a mature, open-source implementation of the Delay/Disruption-Tolerant Networking (DTN) Bundle Protocol (BP), developed primarily for space communications but also applicable to terrestrial challenged networks (11). ION provides a complete DTN node stack including bundle processing, contact plan management, convergence layer protocols (such as TCPCL and LTP), and administrative tools for monitoring and management (15). It has been deployed in NASA missions, satellite testbeds, and research environments, serving as the de facto reference implementation for BP evaluation and experimentation (11, 15).

1.6.1 System Architecture

ION operates as a distributed daemon-based system where the primary `iond` process manages the global DTN node state, bundle storage, and contact scheduling (11). Bundles are persisted in a SQLite database for reliability across restarts, with separate tables tracking bundle metadata, custody records, and delivery status to support proactive custody transfer and retransmission (15). The architecture separates core BP logic from convergence layers via a plugin model, allowing ION to adapt to diverse underlying links including UDP, TCP, LTP, and space data link protocols without modifying the bundle core (11, 16).

1.6.2 Applications and Deployment

ION ships with reference applications demonstrating BP usage, including the Bundle Relay Utility (BRU) for simple store-forward, the DTNperf tool for throughput measurement, and higher-layer protocol shims for email, file transfer, and web services over BP (15). It has been flight-qualified for NASA's Deep Space Network and CubeSat missions, with operational deployments managing terabytes of scientific data across solar system distances (11). The codebase remains actively maintained with contributions from JPL, ESA, and academic partners, supporting both RFC 5050 BPv6 and emerging BPv7 specifications (15, 16).

1.7 Postfix

Postfix is a Mail Transfer Agent (MTA) that handles sending and receiving emails using the Simple Mail Transfer Protocol (SMTP). It is responsible for transferring emails between mail servers and delivering messages to local user mailboxes in a secure and efficient manner.

1.8 Dovecot

Dovecot is a Mail Delivery Agent (MDA) and IMAP/POP3 server that allows users to access and read their emails stored on the mail server. It manages mailbox formats such as Maildir and provides authentication and retrieval services for email clients.

1.9 Thunderbird

Thunderbird is a Mail User Agent (MUA) that provides a graphical interface for users to compose, send, receive, and manage emails. It communicates with Postfix for sending emails via SMTP and with Dovecot for retrieving emails using IMAP or

POP3.

1.10 Virtual Machines (VMs)

Virtual Machines are isolated computing environments that run independent operating systems on a single physical host. In this setup, VMs are used to simulate separate mail servers and clients, enabling realistic testing and experimentation with email system architectures.

1.11 ION DTN

ION-DTN is the reference implementation of Delay-Tolerant Networking protocols by NASA's Jet Propulsion Laboratory, tailored for space communications where standard Internet protocols are unsuitable because of very high latencies, frequent disconnections, and highly asymmetric data rates. ION implements the Bundle Protocol (16), offering store-and-forward message switching for reliable data transport over challenged networks. The protocol uses Contact Graph Routing (CGR) to determine optimal paths for forwarding messages according to expected contact opportunities between nodes in the network, which is particularly useful for interplanetary communications where link schedules are deterministic but discontinuous. ION also supports a variety of convergence layer adapters, including UDP (for testing on Earth), LTP (Licklider Transmission Protocol (17), for deep space communications), and TCP (18), allowing the protocol to be used in a wide range of network environments. The Bundle Security Protocol (19) provides end-to-end authentication and confidentiality for bundles traversing untrusted intermediary nodes. Originally developed for Mars exploration and the Deep Space Network, ION has been successfully tested in space communications and applied to Earth-based scenarios where disruption-tolerant networking is required, such as disaster recovery networks, military communications, and remote sensing.

2 | Literature Review

3 | System Design

System design content

4 | Implementation

Implementation content

5 | Results and Discussion

Results

6 | Conclusions and Future Work

Conclusions

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